EMPIRICAL DURATION

Learning Outcome

- describe the difference between empirical duration and analytical duration

The approaches taken so far to estimate duration and convexity statistics using mathematical formulas is often referred to as **analytical duration**; the measures we have covered are summarized in Exhibit 6. Importantly, estimates of the impact of benchmark yield changes on bond prices using analytical duration implicitly assume that government bond yields and spreads are independent variables and are uncorrelated. Analytical duration offers a reasonable approximation of the price–yield relationship in many situations.

Exhibit 6:

Curve-Based and Empirical Duration Measures

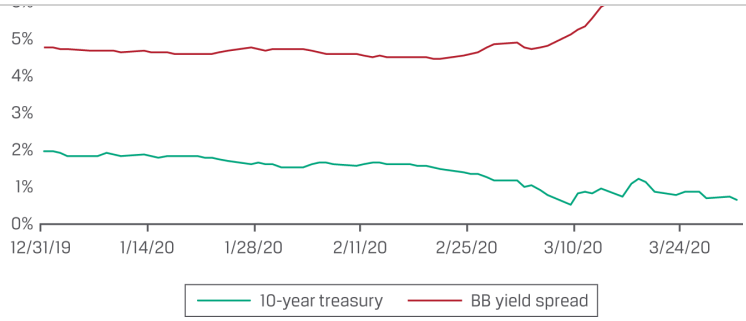
Measure	Definition	Interpretation
Approximate Modified Duration	Estimates the slope of the line tangent to a bond's price–yield curve	Yield-based method to estimate modified duration
Effective Duration	Sensitivity of a bond's price to a change in a benchmark yield curve	Curve-based method to estimate modified duration for complex bonds with uncertain cash flows
Key Rate	Measures bond sensitivity to a benchmark yield change for a specific maturity	Partial duration statistic that gauges a bond's sensitivity to non-parallel benchmark yield curve changes
Empirical	Measure using historical data in statistical models and incorporating factors affecting bond prices to determine the price–yield relationship	Statistical estimate that accounts for correlation between yield spreads and benchmark yield-to-maturity changes under different economic scenarios

However, in practice, there is another important type of duration: Fixed-income professionals often use historical data in statistical models that incorporate various factors affecting bond prices to calculate **empirical duration** estimates (see Exhibit 6). These estimates are calculated over time and in different interest rate environments to inform the fixed-income portfolio decision-making process.

For instance, during crises, investors typically sell risky assets and purchase government bonds. Such a "flight to quality" can cause government benchmark yields to fall while credit spreads widen. Exhibit 7 depicts the flight to quality during the onset of the COVID-19 pandemic at the end of February 2020.

Exhibit 7:

10-Year US Treasury Yield vs. US Corporate BB Credit Spread, Q1 2020



During such periods, we might expect analytical and empirical duration estimates to differ. For a government bond with little or no credit risk, we would expect analytical and empirical duration to be similar because benchmark yield changes largely drive bond prices. Conversely, credit spreads (and liquidity spreads) would widen because of an increase in expected default risk. Since credit spreads and benchmark yields are negatively correlated under this scenario, wider credit spreads will partially or fully offset the decline in government benchmark yields, resulting in lower empirical duration estimates than analytical duration estimates (in other words, the expected *increase* in a bond's price from declining government benchmark yields is offset by a *decrease* in price from widening credit spreads). Analysts must consider the correlation between benchmark yields and credit spreads when deciding whether to use empirical or analytical duration estimates.

EXAMPLE 4

Viswan Family Office's Corporate Bond Portfolio—Empirical vs. Analytical Duration

In addition to equity holdings, the Viswan Family Office (VFO) manages a government bond portfolio and a corporate bond portfolio. Positions in the government bond portfolio are mainly medium-term US Treasury securities but also include debt of other highly rated developed-market sovereign issuers. About half of the corporate bond portfolio is invested in investment-grade issues, and the other half consists of high-yield issues, with a mix of maturities and a mix of North American, European, and Asian companies.

Explain why empirical duration is likely to be a more relevant interest rate risk measure for VFO's corporate bond portfolio than for its government bond portfolio.

Solution:

The government bond portfolio includes debt securities of the US government and other highly rated developed-market sovereign issuers. Since benchmark yields are the primary driver of changes in overall bond yields in this portfolio, the results of analytical duration and empirical duration should be broadly similar.

The corporate bond portfolio includes a wide variety of debt securities with varying levels of credit quality and liquidity risk and, therefore, different credit and liquidity spreads.

Correlations between benchmark yield changes and the sizes of the credit and liquidity spreads are negative, particularly during stressed market conditions, such that during such times, lower benchmark yields are more than offset by higher credit and liquidity spreads. So, for the lower-quality corporate bonds, yields increase, which makes their empirical duration significantly lower than their analytical duration.

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Question Set

1. When comparing analytical duration and empirical duration, which of the following statements is correct?

   A. Empirical duration and convexity are estimated duration and convexity statistics using mathematical formulas.
   B. Analytical duration and convexity are estimated using historical data in non-statistical models that incorporate various factors affecting bond prices.
   C. Neither A nor B

   Hide Solution

   Solution:

   C is correct.

   A is incorrect because it is *analytical*, not empirical, duration and convexity that are measured using mathematical formulas.

   B is incorrect because it is *empirical*, not analytical, duration and convexity that are estimated using historical data in non-statistical models that incorporate various factors affecting bond prices.

2. True or false: Empirical duration estimates provide better estimates than the analytical duration estimates for bonds that have credit risk.

   Hide Solution

   Solution:

   True. Macroeconomic factors drive government bond yields lower in a market stress scenario, and the flight to safety will cause high-yield bond credit spreads to widen because of an increase in expected default risk. Since credit spreads and benchmark yields are negatively correlated under this scenario, wider credit spreads will partially or fully offset the decline in government benchmark yields. Analytical duration, however, assumes that benchmark yields and spreads are uncorrelated. Empirical duration estimates are a more accurate method of forecasting for bonds with credit risk.

3. Identify some of the challenges that may be encountered in estimating empirical duration.

   Hide Solution

   Solution:

   Empirical duration estimation utilizes historical data based on multiple factors. What factors to use and not use, what weight to give the factors, and how to weight current data versus older data (e.g., the time period to measure duration) are some challenges.

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Curve-Based and Empirical Fixed-Income Risk Measures

OPTIONAL UNTESTED SEGMENT

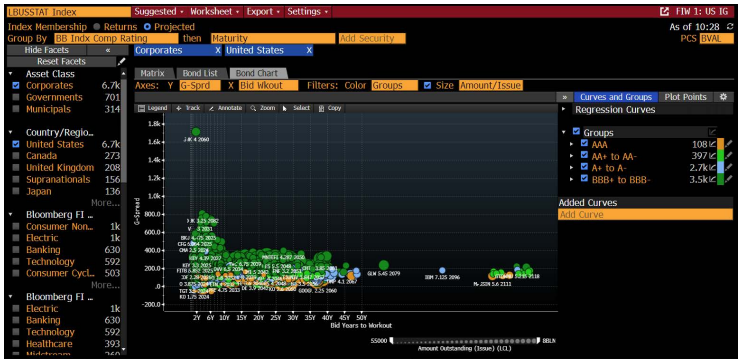
UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application

The Bloomberg FIW function allows you to view how lower-rated bonds have higher spreads over the government curve. You can also understand how longer-tenor bonds tend to have higher spreads than shorter-tenor bonds, given the risk of inflation.

> "FIW <GO> allows you to create and share custom worksheets to analyze pricing, performance, relative value, and liquidity across lists of up to 20,000 bonds. You can analyze lists from multiple Bloomberg Terminal® sources and progressively refine the universe of bonds to discover new investment ideas and opportunities." -Bloomberg

To explore this function, type FIW into the Bloomberg command line and press Enter or <GO>. Select the "United States" checkbox on the left panel. Then click on the gray "Bond Chart" tab to visualize performance for your security list based on specific bond characteristics.



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PJ does empirical use statistical or non statistical model?

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